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import cv2
import numpy as np
from google.colab.patches import cv2_imshow

# Read uploaded image
img = cv2.imread('/content/image.jpg')

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Laplacian mask with positive center coefficient
laplacian_mask = np.array([
    [0, -1, 0],
    [-1, 5, -1],
    [0, -1, 0]
], dtype=np.float32)

# Apply sharpening mask
sharpened = cv2.filter2D(gray, -1, laplacian_mask)

# Display Original Image
print("Original Image")
cv2_imshow(img)

# Display Output Image
print("Sharpened Image")
cv2_imshow(sharpened)
```



Original Image

